

HUNAIN AHMED

AI / Machine Learning & Full-Stack Systems Engineer

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Karachi, Pakistan
Available for Engineering Roles

TECHNICAL SKILLS & CORE COMPETENCIES

CORE // STACK

Languages & Core:	Python, TypeScript, JavaScript (ES+), SQL, C/C++, PHP, HTML5, CSS3 / Tailwind CSS
AI & Machine Learning:	PyTorch, TensorFlow, Scikit-Learn, LangChain, Hugging Face, LLMs, Autonomous Agents, Agent-Based Modeling, Reinforcement Learning, Multi-Modal Gemini API, Model Fine-Tuning, Vector Embeddings, Hybrid RAG, NLP
Computer Vision & Audio:	OpenCV, Computer Vision, Web Audio API (Audio Graph Rendering), Canvas 2D (60 FPS), Edge AI, Timeline Compositing, Data Visualization (Matplotlib, Seaborn, Pandas, NumPy)
Full-Stack & Frameworks:	React 19 / 18, Next.js, React Native, Node.js, NestJS, FastAPI, Laravel, REST APIs, WebSockets, Tkinter, WordPress (CMS & Headless), Zustand, AsyncIO
Databases & Infrastructure:	PostgreSQL (pgvector), Microsoft SQL Server, Redis, PhpMyAdmin, RDBMS Design, Docker, Git, GitHub Actions (CI/CD), Linux/Bash, Mapbox API, Vite
Methodologies & Practices:	Test-Driven Development (TDD), Agile Methodologies, Software Architecture, Debugging & Profiling, Data Cleaning, Analytical Problem Solving

SOFTWARE ENGINEERING & AI SYSTEMS EXPERIENCE

SYSTEMS ARCHITECTURE

Klyptic — Browser-Native Non-Linear Video Editor & AI Subtitle Studio | Lead Architect 2025 – Present

- React 18, TypeScript, HTML5 Canvas 2D, Web Audio API, Tailwind CSS, Zustand
- Architected an in-browser non-linear editing suite delivering zero-server video rendering and sustained 60 FPS compositing via hardware-accelerated Canvas 2D and requestAnimationFrame loops.
 - Engineered non-destructive multi-track sequencing with sub-frame keyframe interpolation, dynamic transition shaders, and audio-video timeline synchronization.
 - Integrated local AI auto-captioning engine with millisecond word-timestamp alignment and offline audio graph rendering via Web Audio API context, reducing backend compute costs to zero.

Trade Bot Gemini — Autonomous Multi-Modal Trading Agent | AI Engineer & Creator 2025 – Present

- Python, Google Gemini Multi-Modal API, NumPy, Pandas, AsyncIO, Autonomous Reasoning Loops
- Engineered an autonomous multi-modal quantitative trading agent ingesting live price action, order-book depth, and macroeconomic sentiment to generate structured, risk-weighted hypotheses.
 - Designed a structured self-auditing reflection loop enforcing strict downside limits, executing decision inference cycles in under 800ms with zero policy breaches.
 - Implemented dynamic volatility calibration and adaptive position sizing based on real-time rolling drawdown thresholds.

MindCare AI — AI Mental Health Therapist & Crisis Support Chatbot | Full-Stack AI Engineer 2024 – 2025

- React.js, FastAPI, Python, AI/NLP, SQLAlchemy, SQLite, JWT Auth, Google OAuth, Geolocation API, Responsive UI
- Developed a full-stack AI mental health support chatbot providing empathetic conversational responses, real-time emotional intent detection, and automated crisis safety checks.
 - Engineered dual Google OAuth and JWT authentication, persistent conversational histories via SQLAlchemy ORM, and location-based recommendation of nearby accredited mental health facilities.

Pakistan Super App — Mobility & Roadside Logistics Platform | Full-Stack Systems Engineer 2024 – 2025

- TypeScript, React Native, Node.js, WebSockets, Mapbox API, Redis, Monorepo
- Architected a cross-platform mobility monorepo unifying passenger, driver, and emergency roadside recovery mobile clients with a scalable Node.js backend service layer.
 - Implemented a sub-second reverse-auction bidding protocol over persistent bidirectional WebSockets with automated geospatial geohash proximity dispatch.

Movie Recommender AI — Vector Similarity & Hybrid ML Engine | ML Developer 2024

- Python, Scikit-Learn, Pandas, NLP Embeddings, Truncated SVD, Cosine Similarity
- Formulated a hybrid recommendation engine synthesizing high-dimensional NLP plot embeddings with sparse collaborative rating matrices to mitigate user cold-start bias.
 - Optimized vector space retrieval using Truncated SVD dimensionality reduction, achieving < 10ms retrieval latency across large candidate catalogs.

Full-Stack & AI Systems Consultancy | Independent Software Developer (Upwork & Global Clients) 2024 – Present

- React, Next.js, Python, PHP, Laravel, WordPress, Docker, PostgreSQL, REST APIs
- Delivered high-performance web applications, custom CMS solutions (WordPress, PHP, Laravel), and headless modern frontends (Next.js/React) for international clients.
 - Implemented custom automated data processing pipelines and robust API endpoints with automated unit testing and containerized Docker deployments.

COMPETITIVE PROGRAMMING & TECHNICAL HONORS

AWARDS & CONTESTS

PROCOTM — National SQL & Database Optimization Contest | FAST-NUCES FAST-NUCES

Competed in intensive database query-solving; authored optimized multi-table joins, subqueries, and analytical SQL scripts under strict timed limits.

SuperComp — Speed Programming & Algorithmic Problem Solving | Bahria University (BUKC) BUKC

Solved complex algorithmic challenges and edge-case test suites in C++ and Python under strict timed conditions, optimizing time and space complexity.

SuperComp — Code in the Dark (Blind Frontend Engineering Contest) | Bahria University (BUKC) BUKC

Competed in high-pressure blind frontend contest: analyzed an intricate target web interface shown briefly, then reproduced the pixel-precise responsive layout from memory using pure HTML/CSS without live browser previews.

TEKNOFEST International Aerospace & Technology Festival | Karachi Expo Center Expo Center

Participated in international technology competition tracks, collaborating on technical problem-solving and innovation-focused engineering solutions.

EDUCATION & FORMAL FOUNDATIONS

ACADEMIC FOUNDATION

Bahria University Karachi — Bachelor of Science in Artificial Intelligence (BS AI) 2024 – Present

Key Coursework: Deep Learning, Neural Networks, Natural Language Processing, Computer Vision, Probabilistic Graphical Models, Data Structures & Algorithms, Database Systems (RDBMS), Systems Architecture.